

Convergence Properties of the Finite Element Solution

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Abstract—When analyzing field problems with the finite element method, the numerical solution will converge against the physical solution. This paper deals with the influence of singularities on the convergence behaviour.

Index terms—Convergence of numerical methods, finite element methods, electrostatic analysis, magnetostatics

I. CONVERGENCE RATE

When examining the convergence of the finite element method, the starting point is the Taylor expansion [1]. This leads to a convergence rate of the order

$$|||e||| = O(h^p) \quad (1)$$

with $|||e|||$ being the energy norm of the error, h the element size, and p the polynomial order of the elements.

For real applications the convergence rate normally is not achieved, since singularities will reduce the convergence rate to

$$|||e||| = O(h^\vartheta), \quad \vartheta = \min(p, \lambda) \quad (2)$$

with λ being the intensity of the singularities [1, 2]. For elasticity problems λ is in the range of 0.5...0.7, depending on the singularity.

Equation (2) describes the behaviour for a uniform change of h with regard to the singularity. If adaptive mesh generation is applied, the convergence rate ϑ is significantly influenced by the chosen refinement [3].

When examining the convergence behaviour, the influence of a singularity cannot be separated from the whole problem. It can only be found the mixture of the singularity and the rest of the region. In this paper different types of singularities and the resulting overall convergence behaviour are examined. For comparison one singularity is combined with a smooth surrounding. The mesh generation is performed without adaptive schemes.

As the element size h in (2) is difficult to define, the number of degrees of freedom (NDF) can be introduced.

As h decreases with $\text{NDF}^{-1/2}$ in the 2d case and $\text{NDF}^{-1/3}$ in the 3d case, the convergence rate ϑ can be calculated from

$$|||e||| = O(\text{NDF}^{-\vartheta/2}) \quad \text{or} \quad |||e||| = O(\text{NDF}^{-\vartheta/3}). \quad (3)$$

For 2d problems ϑ can easily be obtained from (3) by comparing two error values for different NDFs:

$$\vartheta = -2 \frac{\ln |||e_1||| - \ln |||e_2|||}{\ln \text{NDF}_1 - \ln \text{NDF}_2} \quad (4)$$

As a test for the convergence behaviour the (estimated) error can be plotted over the NDF in double logarithmic scale and should give a straight line. Since the absolute value of the error is difficult to interpret, the relative error $\delta = |||e|||/|||u|||$ (with $|||u|||$ being the energy norm of the solution u) or the estimated relative error $\tilde{\delta}$ should be used. Fig. 1 shows the plot for the boundary condition singularity in Fig. 2.

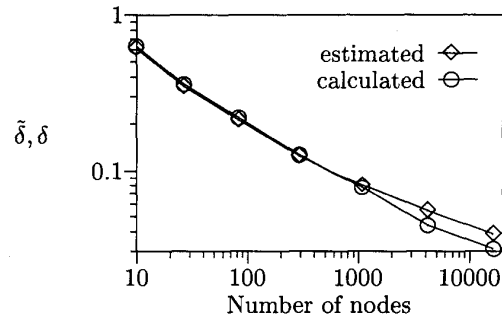


Fig. 1. Convergence for singularity by boundary condition

The investigations have been carried out for the usual case of the Laplace equation, e.g. for electrostatic fields:

$$\Delta u = 0. \quad (5)$$

The error values were obtained using the Bank-Weiser error estimator as described in [3, 4].

II. TYPES OF SINGULARITIES

A. Boundary Conditions

Singularities can occur at points with changing boundary conditions, e.g. from Neumann to Dirichlet bound-

ary condition as presented in [3] (Fig. 2). The convergence rate was $\vartheta = 0.67$. In many cases these changes in the boundary conditions are combined with geometric changes, but these will be treated separately.

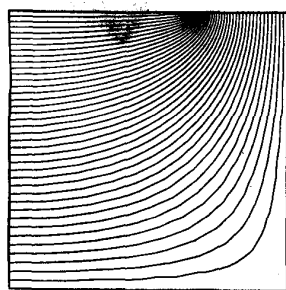


Fig. 2. Singularity by boundary condition

B. Geometric Singularities

Geometric singularities occur at corners. We have investigated the behaviour at two types of corners: corners with natural boundary conditions and corners with imposed boundary conditions.

Both were examined for the case of a corner of varying angle, realized within a circle. The corner is located at a radius of 1, the outer limit of the problem at a radius of 3.

1) *Natural Boundary Condition*: In Fig. 3 a 90 degree corner with Neumann boundary condition on both sides of the corner is shown. The Neumann boundary condition is the natural case which tends to be fulfilled by the solution when the number of degrees of freedom is increased (the mesh size decreased), but it is not fulfilled exactly.

For the case of a 90 degree corner in Fig. 3 we obtained $\vartheta \approx 0.86$. At 179 degrees (Fig. 4) the convergence rate is reduced to $\vartheta \approx 0.66$.

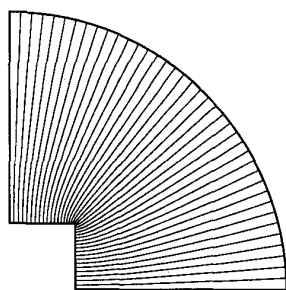


Fig. 3. Singularity at 90 degree corner with natural boundary condition

The convergence behaviour for the different angles yields the expected straight lines within acceptable limits (Fig. 5). Small variations are normal and can occur for example because of the discretization of the surrounding circular boundary. With a small number of elements the

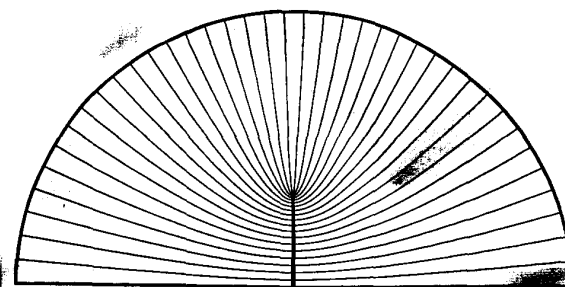


Fig. 4. Singularity at 179 degree corner with natural boundary condition

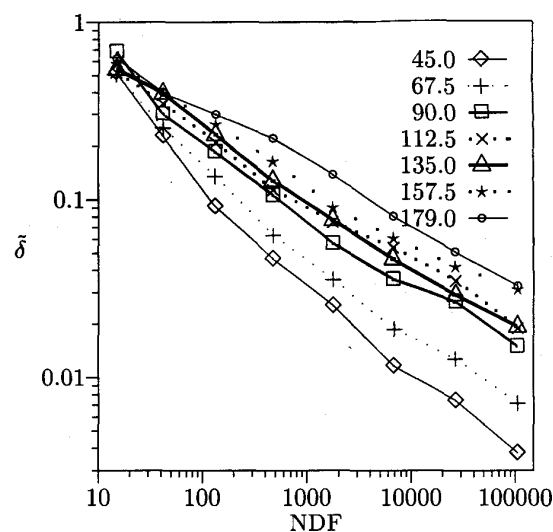


Fig. 5. Convergence of solution for corners with natural boundary condition

boundary is only roughly discretized, but with an increasing number of elements the boundary becomes nearly circular, in fact changing the discrete problem to be solved.

TABLE I
Convergence rate for corner singularity with natural boundary condition

angle	ϑ
45.0	1.109869
67.5	0.960126
90.0	0.861392
112.5	0.746920
135.0	0.752575
157.5	0.658608
179.0	0.664011

The angle of the corner and the convergence rate are listed in Table I. For comparison the values were obtained with (4) from the first value at 15 nodes and the final values at 107457 nodes. All meshes were generated from a very coarse starting mesh by bisectioning all edges with following mesh smoothing including edge swapping and node moving [3]. The mesh quality was high: typical values were 0.928 for the mean quality $\bar{Q}$ and 0.41 for

the quality of the worst element $Q_{\min}$ at the end of the process. The quality $Q = 2 \cdot r/R$ being the ratio of the radii of the inscribed circle and the circumcircle.

2) *Imposed Boundary Condition*: Figure 6 shows the same 90 degree corner, but in this case the corner is created by Dirichlet boundaries. The Dirichlet boundary condition is imposed on the solution and hence fulfilled exactly.

The convergence rate for 90 degrees was $\vartheta \approx 0.79$. At 179 degrees (Fig. 7) the convergence rate is reduced to $\vartheta \approx 0.57$.

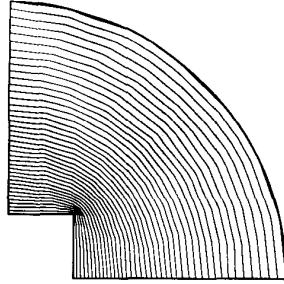


Fig. 6. Singularity at 90 degree corner with imposed boundary condition

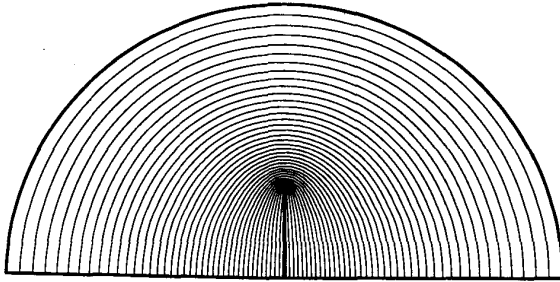


Fig. 7. Singularity at 179 degree corner with imposed boundary condition

Figure 8 shows the convergence of the solution for the different angles of the corner. Again the convergence is smooth within acceptable limits.

The angle of the corner and the convergence rate are listed in Table II.

TABLE II
Convergence rate for corner singularity with imposed boundary condition

angle	ϑ
45.0	0.994248
67.5	0.858107
90.0	0.778797
112.5	0.674015
135.0	0.670405
157.5	0.582329
179.0	0.573969

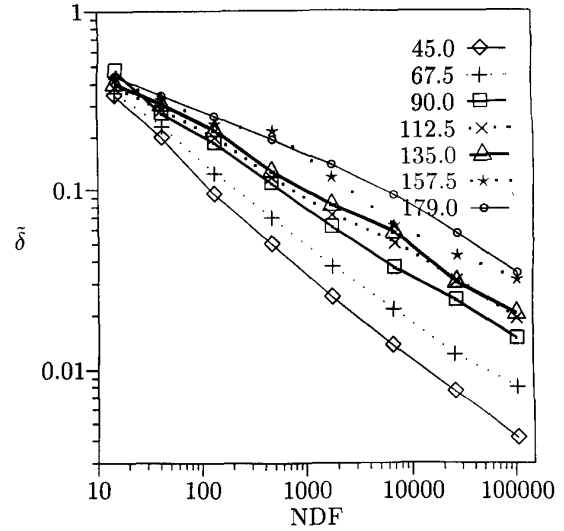


Fig. 8. Convergence of solution for corners with imposed boundary condition

3) *Comparison of geometric singularities*: As can be seen, the imposed boundary condition leads to a stronger intensity of the singularity. For the natural boundary condition the field can violate the boundary condition with only a small "penalty" in the error, but leading to a softening of the singularity. This is not possible for the imposed boundary condition, so that the achieved convergence rate is smaller.

The computed convergence rates are in the same range as the values found for the intensity of singularities in elasticity, but this is no surprise.

4) *Geometric Singularity versus Boundary Condition Singularity*: For the natural boundary condition it takes a very strong geometric singularity (significantly more than 135 degrees) to reduce the convergence rate to the value of $\vartheta \approx 0.67$ that was obtained for the example in Fig. 2. It is however no problem to create a geometric singularity of the same intensity with imposed boundary conditions.

III. COMPLEX GEOMETRY

Figure 9 shows the field in a switched reluctance motor. The calculation was made for a linear model with the relative permeability of the iron being $\mu_r = 1000$. Corners of the iron parts therefore behave similar to geometric singularities. As $\mu \neq \infty$ the intensity of the singularities is however limited. The model is taken from a practical problem, so that the inner iron corners are smoothed and only the corners at the airgap exhibit their sharp geometry. Since the flux lines have to enter the iron nearly perpendicular, these corners should behave like geometric singularities with natural boundary conditions. The angle is roughly 90 degrees, so that a convergence rate of $\vartheta \approx 0.86$ for one singularity should be expected from the

data in Table I.

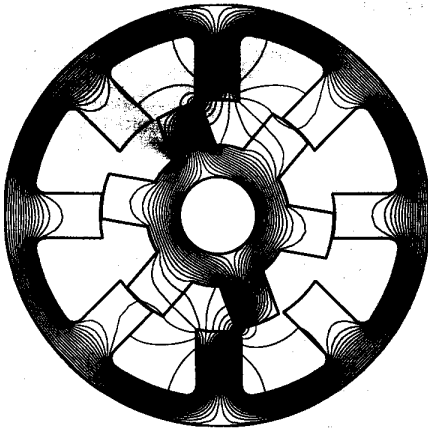


Fig. 9. Switched reluctance motor

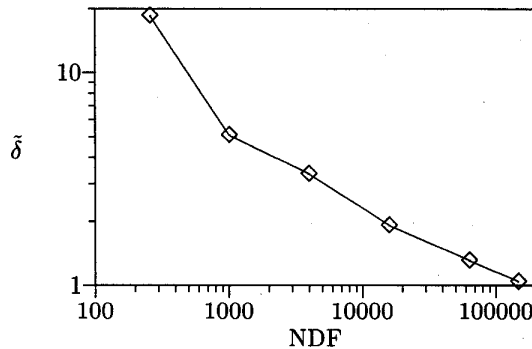


Fig. 10. Convergence of solution for switched reluctance motor

Figure 10 shows the convergence behaviour for the SR motor. While the improvement for the first step is rather high, it should be ignored, because it is related to a vast improvement in the mesh quality (starting with only 260 nodes) and the geometric modelling of the arcs.

The overall convergence rate for this problem, starting with the second step, is $\vartheta = 0.64$. This value is rather low, but there are several iron corners in the problem, each forming a singularity. On the other hand, large regions are nearly flux free and hence the additional unknowns introduced in these regions are wasted. The overall error is dominated by the singular regions, leading to a rather bad ratio of error to NDF.

As a comment it should be noted, that the convergence rate can be improved to $\vartheta \approx 2$ with adaptive mesh generation.

IV. CONCLUSIONS

The influence of different types of singularities has been examined empirically. Geometric singularities exhibit dif-

ferent behaviour when made up of natural or imposed boundary conditions. The intensity is stronger in the case of imposed boundary conditions.

In problems with a larger number of singularities and or flux free regions the convergence rate can become rather bad because the error rating is dominated by the singular regions and additional degrees of freedom introduced into the flux free regions will not improve the solution.

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